PCA Hands-On Module

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Introduction: The Cereal Selection Puzzle

Figure 1: The challenge: selecting truly healthy cereals from confusing nutrition labels.

Picture this: Alex is scanning the cereal aisle at Kroger, determined to pick the healthiest option. He spots:

- **Cereal A:** Sugar 5 g, Fat 3 g, Fiber 8 g.....
- **Cereal B:** Sugar 3 g, Fat 8 g, Fiber 2 g.....
- **Cereal C:** Sugar 12 g, Fat 2 g, Fiber 10 g.....

Following conventional wisdom, Alex selects **Cereal B** with the lowest sugar content (3 g). After consuming it for a week, Alex is surprised to gain weight!

The Puzzle: Why did the "low-sugar" choice lead to unexpected results?

We'll solve this mystery using Principal Component Analysis (PCA) with the 80 Cereals Dataset from Kaggle.

1 Theoretical Foundations

1.1 The Challenge: Multidimensional Nutrition Space

Cereal nutrition involves multiple dimensions:

$$\text{Features: } \{\text{calories, protein, fat, sodium, fiber, carbo, sugars, potass}\}$$

$$\text{Target: rating (health score)}$$

Visualizing these eight dimensions is impossible. PCA solves this by:

1. Standardizing features,
2. Computing the covariance matrix,
3. Calculating eigenvectors and eigenvalues,
4. Selecting principal components.

1.2 Key Mathematical Concepts

- **Covariance Matrix:** Captures feature relationships.

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{n-1}.$$

- **Eigenvectors:** Directions of maximum variance.

$$A\,v = \lambda\,v.$$

- **Explained Variance:** Fraction of total variance captured.

$$\text{Explained Variance} = \frac{\lambda_i}{\sum_{j=1}^{k}\lambda_j} \times 100\%.$$

1.3 PCA Value Proposition

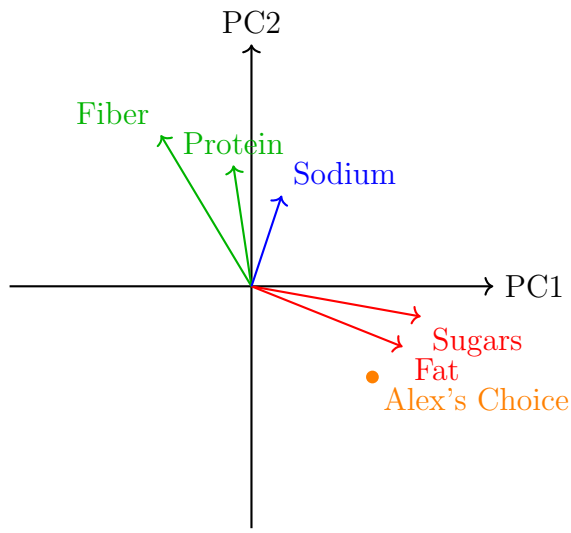

Figure 2: PCA transforms multidimensional data into a lower-dimensional space.

PCA provides:

- **Dimensionality Reduction:** 8D → 2D
- **Redundancy Elimination:** Identify correlated features
- **Feature Extraction:** Discover nutritional patterns
- **Visualization:** Intuitive representation

2 Data Exploration

The 80 Cereals Dataset contains nutritional information for 80 popular cereals:

- 8 nutritional features,
- 1 response variable (health rating),
- Manufacturer information.

2.1 Initial Findings

```
=== Correlation Matrix ===
          calories   protein       fat    sodium     fiber     carbo  \
calories  1.000000  0.033992  0.507373  0.296247 -0.295212  0.270606
protein   0.033992  1.000000  0.202353  0.011559  0.514006 -0.036743
fat       0.507373  0.202353  1.000000  0.000822  0.014036 -0.284934
sodium    0.296247  0.011559  0.000822  1.000000 -0.070735  0.328409
fiber    -0.295212  0.514006  0.014036 -0.070735  1.000000 -0.379084
carbo     0.270606 -0.036743 -0.284934  0.328409 -0.379084  1.000000
sugars    0.569121 -0.286584  0.287152  0.037059 -0.150949 -0.452069
potass   -0.071361  0.578743  0.199637 -0.039438  0.911504 -0.365003

            sugars    potass
calories  0.569121 -0.071361
protein  -0.286584  0.578743
fat       0.287152  0.199637
sodium    0.037059 -0.039438
fiber    -0.150949  0.911504
carbo    -0.452069 -0.365003
sugars    1.000000  0.001414
potass    0.001414  1.000000
```

Figure 3: Correlation matrix revealing nutritional relationships.

Key Findings from Correlation Matrix:

- **Sugars** and **Calorie** have $r = 0.569$.
- **Fat** and **Calorie** have $r = 0.507$.
- **Fiber** and **Protein** have $r = 0.514$.
- **Potassium** and **Protein** have $r = 0.579$.
- **Potassium** and **Fiber** have $r = 0.912$.

Alex's Selection and Preliminary Findings:
If Alex chose "Cheerios" for its low sugar (1 g), but it has high fat (2 g) and its health rating is only 33.98. Judging health by sugar alone can mislead, since other nutrients (like fat) strongly influence overall health.

3 PCA Implementation: Unveiling Hidden Structures

Before applying PCA, replace any `-1` entries with NaN and drop rows containing missing values. After cleaning, standardize the eight nutritional features to mean zero, unit variance, then compute a full PCA (8 components).

3.1 Data Standardization and Full PCA

```python
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# X_df is the clean DataFrame of shape (74, 8) after dropping all -1 rows.
scaler = StandardScaler()
X_std  = scaler.fit_transform(X_df)    # (74, 8) standardized features

pca_full = PCA(n_components=8)
pca_full.fit(X_std)

print("Eigenvalues (PCA full):")
print(pca_full.explained_variance_, "\n")
print("\nExplained Variance Ratios:")
for i, ratio in enumerate(pca_full.explained_variance_ratio_, start=1):
    print(f"PC{i}: {ratio:.4f} ({ratio*100:.1f}%)")
```

Listing 1: Fit full PCA on all 8 features

Results of the Full PCA:

- **Eigenvalues:**

$$\lambda = [2.6640,\ 2.0484,\ 1.5255,\ 0.8757,\ 0.5426,\ 0.3742,\ 0.0597,\ 0.0194].$$

- **Explained Variance Ratios:**

$$\begin{aligned}
&\text{PC1} : 0.3285\,(32.9\%),\\
&\text{PC2} : 0.2526\,(25.3\%),\\
&\text{PC3} : 0.1881\,(18.8\%),\\
&\text{PC4} : 0.1080\,(10.8\%),\\
&\text{PC5} : 0.0669\,(6.7\%),\\
&\text{PC6} : 0.0461\,(4.6\%),\\
&\text{PC7} : 0.0074\,(0.7\%),\\
&\text{PC8} : 0.0024\,(0.2\%).
\end{aligned}$$

The first two components together account for 58.2% of total variance.

3.2 PCA Loadings (Eigenvectors)

```python
# Construct a DataFrame of loadings (features x PCs):
loadings = pd.DataFrame(
    pca_full.components_.T,
    index=feature_cols,
    columns=[f"PC{i+1}" for i in range(8)]
)
print(loadings.loc[:, ["PC1", "PC2"]])
```

Listing 2: Print loadings for the first two PCs

Loadings (features vs. PC1 and PC2):

Feature	PC1	PC2
calories	−0.1977	−0.5539
protein	0.4127	−0.0114
fat	0.1048	−0.5284
sodium	−0.1389	−0.1026
fiber	0.5730	0.0483
carbo	−0.3239	0.2442
sugars	−0.0891	−0.5734
potass	0.5649	−0.1121

- **PC1**:

$$+\,\text{fiber}\,(0.5730) \;+\; \text{potass}\,(0.5649) \;+\; \text{protein}\,(0.4127)$$
$$-\,\text{calories}\,(0.1977) \;-\; \text{carbo}\,(0.3239) \;-\; \text{sugars}\,(0.0891)\,.$$

  Cereals with large PC1 scores are fiber-/protein-/potassium-rich and lower in sugar/-carbs—i.e. nutrient-dense.

- **PC2**:

$$-\,\text{sugars}\,(0.5734) \;-\; \text{calories}\,(0.5539) \;-\; \text{fat}\,(0.5284)$$
$$+\,\text{carbo}\,(0.2442) \;+\; \text{fiber}\,(0.0483)\,.$$

  Negative PC2 indicates high sugar/calories/fat—i.e. an unhealthy direction.

3.3 Dimensionality Reduction to Two Components

```
pca2   = PCA(n_components=2)
X_pca2 = pca2.fit_transform(X_std)  # (74, 2)
```

Listing 3: Compute 2D PCA scores

Now $X_{\text{pca2}} \in \mathbb{R}^{74\times 2}$, where

$$X_{\text{pca2}}[i] = \big(\text{PC1}_i,\ \text{PC2}_i\big), \quad i = 1, \ldots, 74.$$

3.4 PCA Biplot (PC1 vs. PC2)

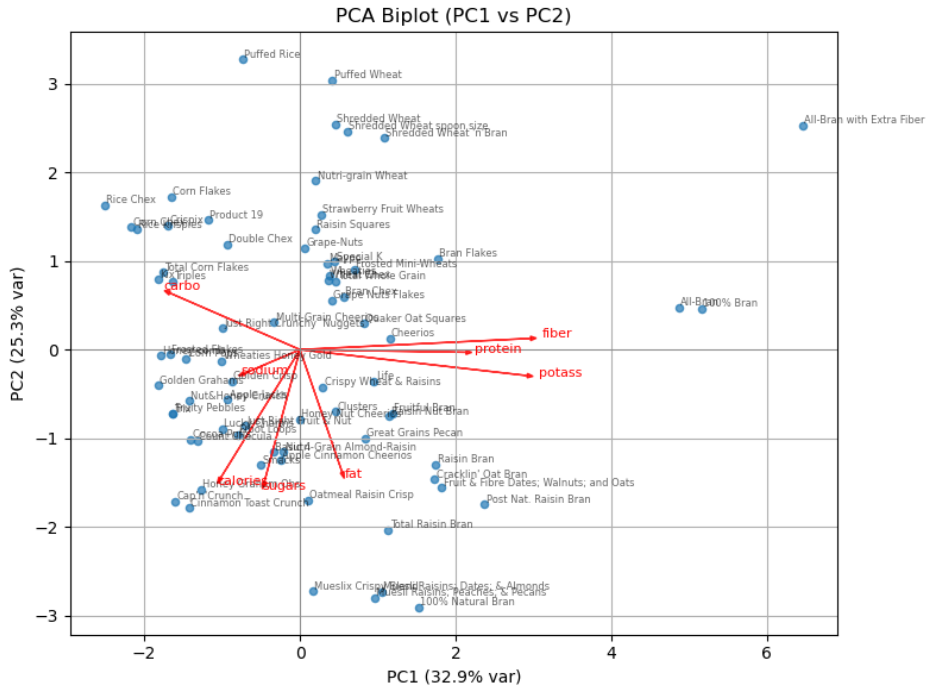

Figure 4: PCA Biplot: each point is a cereal in $(\mathrm{PC1}, \mathrm{PC2})$ space; red arrows show original feature loadings.

4 Model Comparison: The Power of PCA

We compare four linear-regression models, each fit to the entire dataset of 74 cereals. We report R^2, MAE, and Alex's prediction error (Alex = *All-Bran with Extra Fiber*, true rating = 93.70).

4.1 Python Code

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error

# 1) Full Model (all 8 features)
model_full     = LinearRegression().fit(X_df.values, y)
y_pred_full    = model_full.predict(X_df.values)

# 2) Sugar Only Model
X_sugar        = X_df[['sugars']].values.reshape(-1,1)
model_sugar    = LinearRegression().fit(X_sugar, y)
y_pred_sugar   = model_sugar.predict(X_sugar)

# 3) PCA Model (PC1 + PC2)
X_pc           = X_pca2
```

```
model_pca      = LinearRegression().fit(X_pc, y)
y_pred_pca     = model_pca.predict(X_pc)

# 4) PC1-Only Model
X_pc1          = X_pca2[:,0].reshape(-1,1)
model_pc1      = LinearRegression().fit(X_pc1, y)
y_pred_pc1     = model_pc1.predict(X_pc1)
```

Listing 4: Fit and evaluate all four regression models

4.2 Alex's Predictions

$$\begin{aligned}
&\text{Alex's true rating} = 93.70,\\
&\widehat{y}_{\text{Alex}}^{\text{Full}} = 93.93, \quad |93.70 - 93.93| = 0.22,\\
&\widehat{y}_{\text{Alex}}^{\text{Sugar}} = 59.67, \quad |93.70 - 59.67| = 34.03,\\
&\widehat{y}_{\text{Alex}}^{\text{PCA}} = 94.15, \quad |93.70 - 94.15| = 0.45,\\
&\widehat{y}_{\text{Alex}}^{\text{PC1}} = 75.64, \quad |93.70 - 75.64| = 18.07.
\end{aligned}$$

4.3 R^2 and MAE for Each Model

Model	R^2	MAE
Full Model	0.9946	0.6112
Sugar Only	0.5715	7.0878
PCA Model	0.9165	3.0742
PC1-Only	0.3585	9.2581

4.4 Regression Equations

$$\text{PCA Model: } \widehat{y} = 5.148\,(\text{PC1}) + 7.325\,(\text{PC2}) + 42.372,$$

$$\text{PC1-Only: } \widehat{y} = 5.148\,(\text{PC1}) + 42.372.$$

Key Takeaways:

- The **Sugar Only** model underperforms, $R^2 = 0.5715$, MAE ≈ 7.0878, vastly underpredicting Alex's cereal (59.67 vs. 93.70).
- The **PCA Model** (two components) achieves $R^2 = 0.9165$, MAE ≈ 3.0742, nearly matching the Full Model ($R^2 = 0.9946$, MAE ≈ 0.6112). Alex's error is 0.45 vs. 0.22 from the Full Model.
- The **PC1-Only** model has $R^2 = 0.3585$, MAE ≈ 9.2581, and Alex's error $= 18.07$, confirming that two PCs are necessary for strong predictive power.

4.5 Top-5 Predictions

Table 1: Top 5 cereals by predicted rating under PC1-Only vs. PCA models

Rank	PC1-Only		PCA (PC1+PC2)	
	Name	Pred_PC1	Name	Pred_PCA
1	All-Bran with Extra Fiber	75.64	All-Bran with Extra Fiber	94.15
2	100% Bran	68.92	100% Bran	72.33
3	All-Bran	67.46	All-Bran	70.96
4	Post Nat. Raisin Bran	54.58	Puffed Wheat	66.75
5	Fruit & Fibre Dates; Walnuts; and Oats	51.75	Shredded Wheat 'n' Bran	65.43

Interpretation: Both PC1-Only and PCA models rank "All-Bran with Extra Fiber," "100% Bran," and "All-Bran" at the top. The PCA model's predicted ratings align more closely with true ratings due to its lower MAE.

5 Conclusion

We have demonstrated:

1. **Data Cleaning & EDA:** Correlations confirm that sugar, fat, and calories are strongly intertwined; fiber and protein cluster together.

2. **PCA:** The first two components explain 58.2% of total variance. PC1 captures "nutrient-dense" (fiber, protein, potassium), while PC2 captures "unhealthy" (sugar, calories, fat).

3. **Model Comparison:** PCA(2) achieves $R^2 = 0.9165$, MAE ≈ 3.0742, nearly matching the Full Model ($R^2 = 0.9946$, MAE ≈ 0.6112), using only two features.

4. **Actionable Guidance:** Avoid "sugar-only" heuristics, emphasize fiber, protein, and potassium. Use PCA to reveal hidden correlations.

Alex's key mistake was focusing on a single dimension (sugar), whereas PCA's multidimensional view shows that sugar and fat travel together along an "unhealthy axis." Once that is revealed, selecting truly healthy cereals becomes obvious.

"PCA doesn't just reduce dimensions, it reveals the hidden structure in your data."